Experiment No: 8

Aim: To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory:
Ganache is a personal Ethereum blockchain used for testing and development. It allows developers to deploy contracts, develop decentralized applications (DApps), and run tests in a safe and deterministic environment.

Ganache provides:

- Instant mining of blocks
- Pre-funded test accounts
- Transaction history view
- Graphical interface to monitor blockchain activities

It is part of the Truffle Suite and is commonly used with Remix and Metamask for smart contract deployment.

Steps to Connecat Ganache with Metamask and Remix IDE:

1. Install Ganache:
    - Download and install Ganache from the Truffle Suite website.
    - Launch it to start a local blockchain network with test Ether accounts.
2. Connect Ganache Accounts with Metamask:
    - Open Metamask and switch to a Custom RPC Network.
    - Enter the Ganache RPC server URL (e.g., HTTP://127.0.0.1:7545).
    - Import private keys of Ganache accounts into Metamask.
3. Connect Remix IDE with Metamask:
    - Open Remix IDE in the browser.
      Choose Injected Web3 as the environment in the "Deploy & Run Transactions" tab.
    - Remix will automatically connect to the Metamask account (linked with Ganache).
4. Create a Simple Solidity Smart Contract:
    - Write a Solidity program related to your mini project (e.g., Student Record, Donation Tracker, etc.).

```
pragma solidity ^0.8.0;
    contract SimpleStorage {
        uint public data;
        function set(uint x) public {
            data = x;
        }
```

5. Compile and deploy the contract:
   Compile using Remix.
   - Deploy via the Metamask account connected to Ganache.
   - Approve the transaction in Metamask.
6. Check Transaction Details on Ganache:
   - View mined transactions, block details, and balances in the Ganache interface.
7. Interact with the Smart Contract:
   - Use Remix to call contract functions (set() or get()).
   - Observe the state changes and corresponding transactions on Ganache.

Outcome**:**

- Understood how to link **Metamask** with **Remix IDE** for Ethereum smart contract deployment.
- Successfully deployed and interacted with a Solidity smart contract through **Ganache**.
- Verified all transactions and balances within the Ganache environment.

Implementation:

- FarmerContract.sol:

```
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
contract FarmerContract {
   struct Batch {
      uint256 batchID;
      string cropName;
      uint256 quantity;
      uint256 timestamp;
      address farmer;
   }
   mapping(uint256 => Batch) public batches;
   event BatchCreated(
      uint256 batchID,
      string cropName,
      uint256 quantity,
      uint256 timestamp,
      address farmer
   );
   function addBatch(uint256 _batchID, string
memory _cropName, uint256 _quantity) public {
      require(_batchID != 0, "Invalid batch ID");
      require(batches[_batchID].batchID == 0,
"Batch already exists!");

      batches[_batchID] = Batch({
         batchID: _batchID,
         cropName: _cropName,
         quantity: _quantity,
         timestamp: block.timestamp,
         farmer: msg.sender
      });
      emit BatchCreated(_batchID, _cropName,
_quantity, block.timestamp, msg.sender);
   }
   function getBatch(uint256 _batchID) public
view returns (
      uint256,
      string memory,
      uint256,
      uint256,
      address
   ) {
      Batch memory b = batches[_batchID];
      require(b.batchID != 0, "Batch does not
exist!");
      return (b.batchID, b.cropName, b.quantity,
b.timestamp, b.farmer);
   }
}
```

- DistributorContract.sol:

```
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IFarmerContract {
    function getBatch(uint256 _batchID) external
view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address
    );
}
contract DistributorContract {
    struct ReceivedBatch {
        uint256 batchID;
        string cropName;
        uint256 quantity;
        uint256 timestamp;
        address farmer;
        address distributor;
    }
    mapping(uint256 => ReceivedBatch) public
receivedBatches;
    IFarmerContract public farmerContract;
    event BatchReceived(
        uint256 batchID,
        string cropName,
        uint256 quantity,
        address farmer,
        address distributor,
        uint256 timestamp
    );
    constructor(address _farmerContractAddress)
{
        farmerContract =
IFarmerContract(_farmerContractAddress);
    }
    function receiveBatch(uint256 _batchID)
public {
        (
            uint256 id,
            string memory cropName,
            uint256 quantity,
            , // ignore farmer timestamp
            address farmer
        ) = farmerContract.getBatch(_batchID);
        require(id != 0, "Batch does not exist in
FarmerContract");
        require(receivedBatches[_batchID].batchID
== 0, "Batch already received!");
        receivedBatches[_batchID] =
ReceivedBatch({
            batchID: id,
            cropName: cropName,
            quantity: quantity,
            timestamp: block.timestamp,
            farmer: farmer,
            distributor: msg.sender
        });
        emit BatchReceived(id, cropName,
quantity, farmer, msg.sender, block.timestamp);
    }
    function getReceivedBatch(uint256 _batchID)
public view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address,
        address
    ) {
        ReceivedBatch memory rb =
receivedBatches[_batchID];
        require(rb.batchID != 0, "Batch not
received!");
        return (
            rb.batchID,
            rb.cropName,
            rb.quantity,
            rb.timestamp,
```

```
        rb.farmer,
        rb.distributor
```

- RetailerContract.sol:

```
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IDistributorContract {
    function getReceivedBatch(uint256 _batchID)
external view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address,
        address
    );
}
contract RetailerContract {
    struct RetailBatch {
        uint256 batchID;
        string cropName;
        uint256 quantity;
        uint256 timestamp;
        address farmer;
        address distributor;
        address retailer;
    }
    mapping(uint256 => RetailBatch) public
retailBatches;
    IDistributorContract public
distributorContract;
    event BatchReceivedAtRetailer(
        uint256 batchID,
        string cropName,
        uint256 quantity,
        address farmer,
        address distributor,
        address retailer,
        uint256 timestamp
    );
    constructor(address
_distributorContractAddress) {
        distributorContract =
IDistributorContract(_distributorContractAddres
s);
    }

    function receiveFromDistributor(uint256
_batchID) public {
        (
            uint256 id,
            string memory cropName,
            uint256 quantity,
            , // ignore distributor timestamp
            address farmer,
            address distributor
        ) =
distributorContract.getReceivedBatch(_batchID)
;
        require(id != 0, "Batch does not exist in
DistributorContract");
        require(retailBatches[_batchID].batchID ==
0, "Batch already received at Retailer!");
        retailBatches[_batchID] = RetailBatch({
            batchID: id,
            cropName: cropName,
            quantity: quantity,
            timestamp: block.timestamp,
            farmer: farmer,
            distributor: distributor,
            retailer: msg.sender
        });
        emit BatchReceivedAtRetailer(
            id,
            cropName,
            quantity,
            farmer,
            distributor,
            msg.sender,
            block.timestamp
        );
        );
    }
}
```

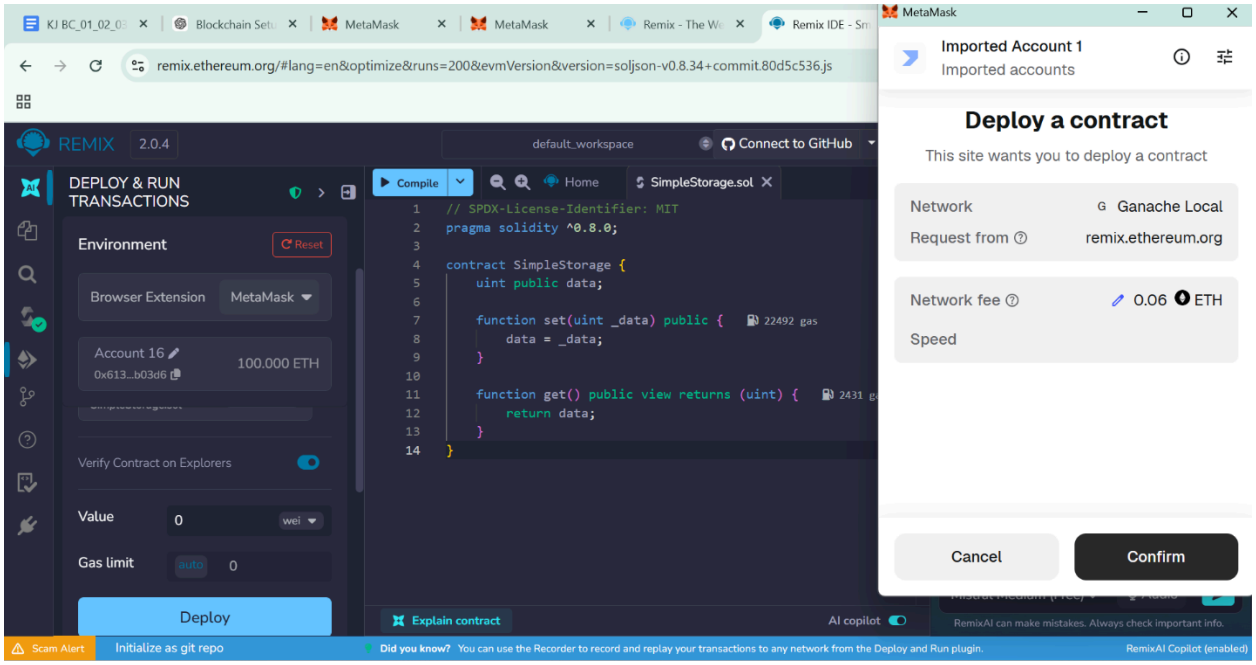
```
    }
    function getRetailBatch(uint256 _batchID)
public view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address,
        address,
        address
    ) {
        RetailBatch memory rb =
retailBatches[_batchID];
        require(rb.batchID != 0, "Batch not
received at retailer!");

        return (
            rb.batchID,
            rb.cropName,
            rb.quantity,
            rb.timestamp,
            rb.farmer,
            rb.distributor,
            rb.retailer
        );
    }
}
```

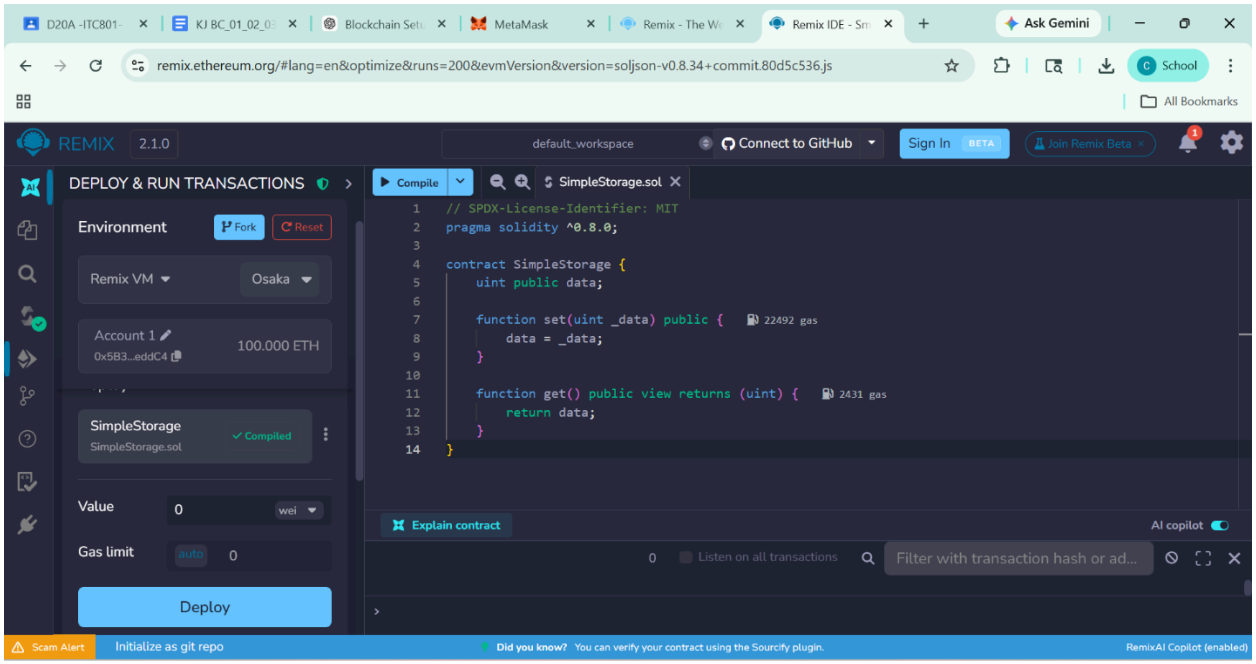
- CustomerContract.sol:

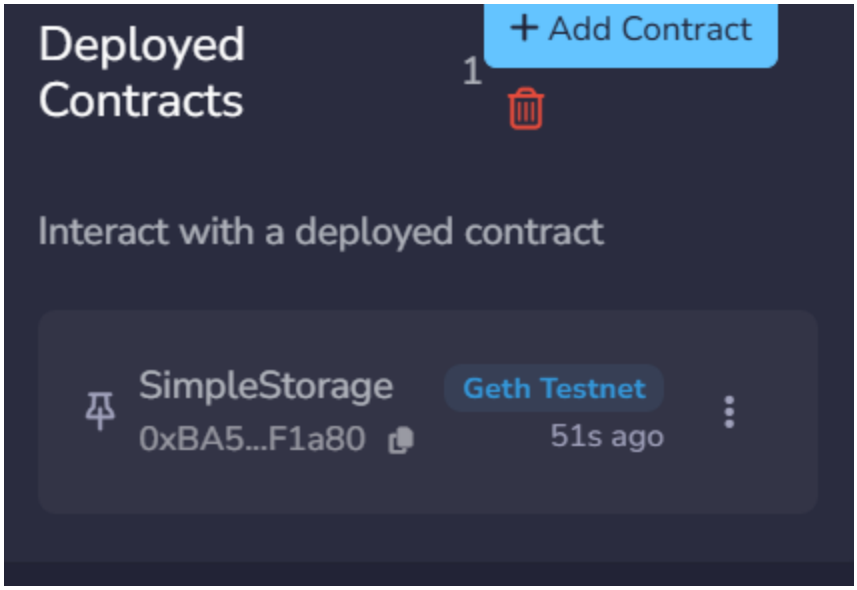
```
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.0;
interface IRetailerContract {
    function getRetailBatch(uint256 _batchID)
external view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address,
        address,
        address
    );
}
contract CustomerContract {
    struct PurchasedBatch {
        uint256 batchID;
        string cropName;
        uint256 quantity;
        uint256 timestamp;
        address farmer;
        address distributor;
        address retailer;
        address customer;
    }
    mapping(uint256 => PurchasedBatch) public
purchasedBatches;
    IRetailerContract public retailerContract;
    event BatchPurchased(
        uint256 batchID,
        string cropName,
        uint256 quantity,
        address farmer,
        address distributor,
        address retailer,
        address customer,
        uint256 timestamp
    );
    constructor(address _retailerContractAddress)
{
        retailerContract =
IRetailerContract(_retailerContractAddress);
    }
    function purchaseFromRetailer(uint256
_batchID) public {
        (
            uint256 id,
            string memory cropName,
            uint256 quantity,
            , // ignore retailer timestamp
            address farmer,
            address distributor,
            address retailer
```

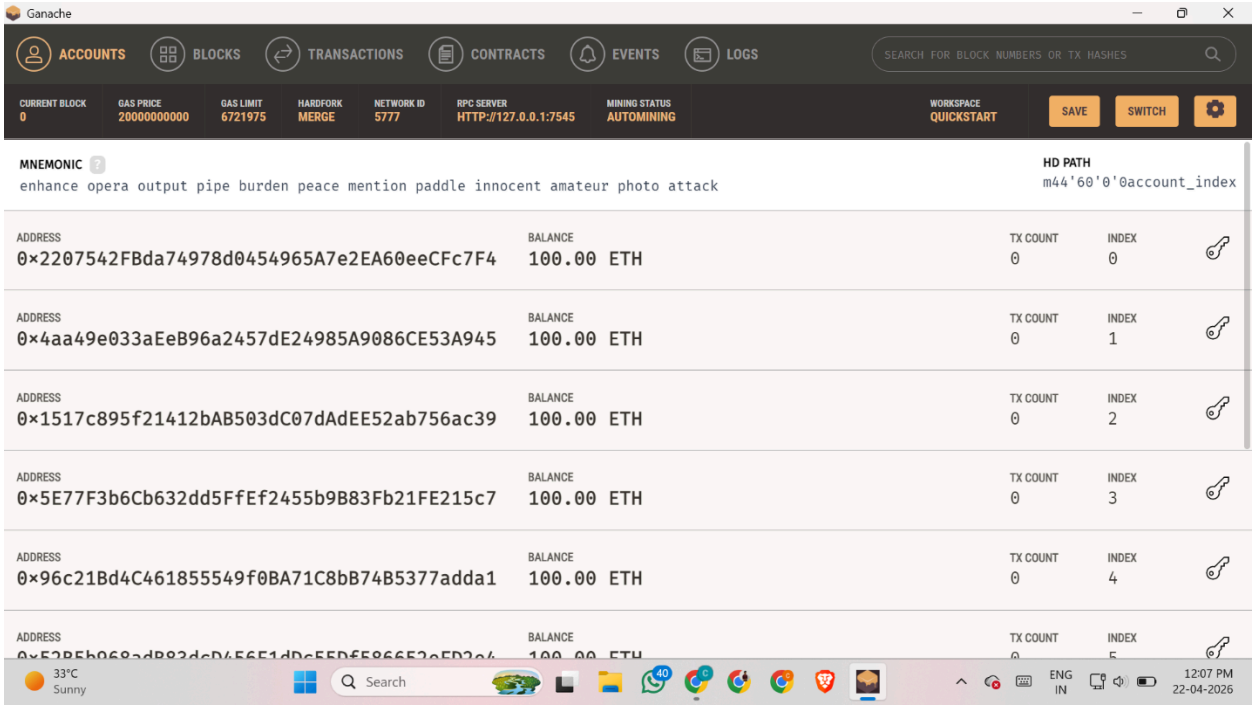
```
        ) =
retailerContract.getRetailBatch(_batchID);
        require(id != 0, "Batch does not exist in
RetailerContract");
    require(purchasedBatches[_batchID].batchID
== 0, "Batch already purchased!");
        purchasedBatches[_batchID] =
PurchasedBatch({
            batchID: id,
            cropName: cropName,
            quantity: quantity,
            timestamp: block.timestamp,
            farmer: farmer,
            distributor: distributor,
            retailer: retailer,
            customer: msg.sender
        });
        emit BatchPurchased(
            id,
            cropName,
            quantity,
            farmer,
            distributor,
            retailer,
            msg.sender,
            block.timestamp
        );
    }

    function getPurchasedBatch(uint256
_batchID) public view returns (
        uint256,
        string memory,
        uint256,
        uint256,
        address,
        address,
        address,
        address
    ) {
        PurchasedBatch memory pb =
purchasedBatches[_batchID];
        require(pb.batchID != 0, "Batch not
purchased!");
        return (
            pb.batchID,
            pb.cropName,
            pb.quantity,
            pb.timestamp,
            pb.farmer,
            pb.distributor,
            pb.retailer,
            pb.customer
        );
    }
}
```

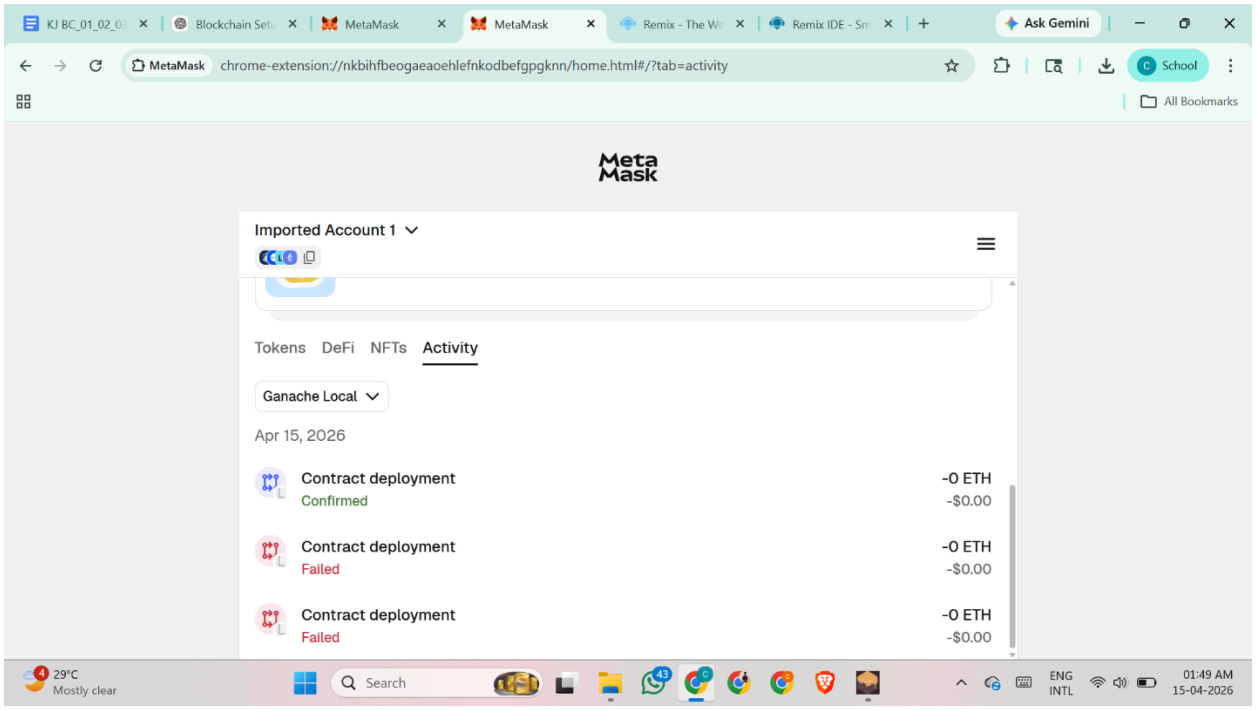
Output:


remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
REMIX 2.0.4
DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
MetaMask
Account 16
0x613...b03d6
100.000 ETH
Verify Contract on Explorers
Value
0
wei
Gas limit
auto
0
Deploy
Compile
Home
SimpleStorage.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract SimpleStorage {
uint public data;
function set(uint _data) public {
22492 gas
data = _data;
function get() public view returns (uint) {
return data;
Explain contract
AI copilot
Scam Alert
Initialize as git repo
Did you know? You can use the Recorder to record and replay your transactions to any network from the Deploy and Run plugin.
RemixAI Copilot (enabled)
MetaMask
Imported Account 1
Imported accounts
Deploy a contract
This site wants you to deploy a contract
Network
Ganache Local
Request from
remix.ethereum.org
Network fee
0.06 ETH
Speed
Cancel
Confirm


remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
REMIX 2.1.0
default_workspace
Connect to GitHub
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Fork
Reset
Remix VM
Osaka
Account 1
0x5B3...eddC4
100.000 ETH
SimpleStorage
SimpleStorage.sol
Compiled
Value
0
wei
Gas limit
auto
0
Deploy
Compile
SimpleStorage.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract SimpleStorage {
uint public data;
function set(uint _data) public {
22492 gas
data = _data;
function get() public view returns (uint) {
2431 gas
return data;
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or ad...
Scam Alert
Initialize as git repo
Did you know? You can verify your contract using the Sourcify plugin.
RemixAI Copilot (enabled)


Deployed Contracts
1
+ Add Contract
Interact with a deployed contract
SimpleStorage
0xBA5...F1a80
Geth Testnet
51s ago


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 0
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
MNEMONIC
enhance opera output pipe burden peace mention paddle innocent amateur photo attack
HD PATH
m44'60'0'0account_index
ADDRESS 0x2207542FBda74978d0454965A7e2EA60eeCFc7F4 BALANCE 100.00 ETH TX COUNT 0 INDEX 0
ADDRESS 0x4aa49e033aEeB96a2457dE24985A9086CE53A945 BALANCE 100.00 ETH TX COUNT 0 INDEX 1
ADDRESS 0x1517c895f21412bAB503dC07dAdEE52ab756ac39 BALANCE 100.00 ETH TX COUNT 0 INDEX 2
ADDRESS 0x5E77F3b6Cb632dd5FfEf2455b9B83Fb21FE215c7 BALANCE 100.00 ETH TX COUNT 0 INDEX 3
ADDRESS 0x96c21Bd4C461855549f0BA71C8bB74B5377adda1 BALANCE 100.00 ETH TX COUNT 0 INDEX 4
33°C Sunny
Search
ENG IN
12:07 PM 22-04-2026



Conclusion:

Ganache provides a simple and powerful environment to simulate a local Ethereum blockchain. By integrating it with **Metamask** and **Remix**, developers can deploy, test, and debug smart contracts efficiently before moving to a public network.